

Task 1. Handling Events in React

- a) Create a simple component with a button labeled “Toggle Message.”
- b) When the button is clicked, toggle a message on the screen that says “Hello, welcome to React!” This message should disappear when the button is clicked again.

Please find the screenshot below for the button component.

- It contains a button named ‘Toggle Message’, with ‘OnClick’ event.
- When we click on ‘Toggle Message’ button, it toggles (hide/ show) a message as in below screenshot.
- This component is called from App.jsx file. A prop is passed for the name to be displayed in the component.
- Thus, when the button is clicked, it displays message with name sent as prop to the component.
- When the button is clicked again, it hides the message.

MessageButton.jsx:

```
MessageButton.jsx U X
Module 8 - Wk10 > Module 8_Week 10_Required Assignment_Patil > mytasks-app > src > Components > MessageButton.jsx > MessageButton
1  import React, { useState } from "react"
2  import './App.css'
3
4  function MessageButton(props){
5      const [showMessage, setShowMessage]=useState(false)
6
7      const handleToggleMessage = () =>{
8          setShowMessage(!showMessage)
9      }
10
11     return(
12         <div>
13             <button title="Toggle Message"
14                 className="button"
15                 onClick={handleToggleMessage}>
16                 Toggle Message
17             </button>
18
19             {showMessage && <h3>Hello {props.name}, welcome to React!</h3>}
20         </div>
21     );
22 }
23
24
25
26 export default MessageButton
```

App.jsx:

```
App.jsx U X
Module 8 - Wk10 > Module 8_Week 10_Required Assignment_Patil > mytasks-app > src > App.jsx > App
1  import { useState } from 'react'
2  import MessageButton from './Components/MessageButton'
3  import ColorChanger from './Components/ColorChanger'
4  import UserProfile from './Components/UserProfile'
5  import './App.css'
6
7  function App() {
8    const [count, setCount] = useState(0)
9
10   return (
11     <>
12       <h1>Task 1 - Handling Events in React</h1>
13       <div>
14         <MessageButton name="Alice" />
15       </div>
16       <h1>Task 2 - State Management using React Hooks</h1>
17       <div>
18         <ColorChanger/>
19       </div>
20       <h1>Task 3 - Asynchronous Data Fetching with AJAX</h1>
21       <div>
22         <UserProfile />
23       </div>
24     </>
25   )
26 }
27
28 export default App
29
```

When button clicked, the message is shown:



Task 1 - Handling Events in React

Toggle Message

Hello Alice, welcome to React!

When the button is re-clicked, the message is hidden



Task 1 - Handling Events in React

Toggle Message

Task 2. State Management using React Hooks

- Build a component called ColorChanger. This component should have a text input where users can enter a colour name (e.g., “blue”).
- The component should display a box that changes colour based on the input.
- As the user types into the input, the box should automatically update to the new colour if it's a valid colour name.

Please find screenshots below for the ColorChanger component.

- It has a textbox for user to enter the name of the color, he/she wants to set for the div box below.
- When the user types the color name into the input box, the background color of the div box gets updated automatically.
- If the user inputs invalid color name, the background color of the div box is set to transparent and the error message is displayed below the div box.
- This component is called from App.jsx file.

ColorChanger.jsx:

```
ColorChanger.jsx U X
Module 8 - Wk10 > Module 8_Week 10_Required Assignment_Patil > mytasks-app > src > Components > ColorChanger.jsx > default
1  import React, { useState } from "react";
2
3  function ColorChanger(){
4      const [colorname, setColorname]=useState('')
5
6      const ChangeColor = (e) => {
7          setColorname(e.target.value)
8      }
9
10     const IsValidColor = () =>{
11         const temp = new Option().style;
12         temp.color=colorname;
13         return temp.color !== '';
14     }
15
16     const isValid = IsValidColor();
17
18     return(
19         <div style={{padding:'20px'}}>
20             <div>
21                 <label title="Enter the background color for the box"
22                     htmlFor="txtColorName">
23                     Enter the background color for the box:
24                 </label>
25                 <input type="text" id="txtColorName" className="textbox"
26                     placeholder="Enter color name e.g. Red..." value={colorname}
27                     onChange={ChangeColor}>
28             </div>
29             <div style={{width:'100px', height:'100px',
30                 border:'1px solid #000000',
31                 backgroundColor: IsValidColor() ? colorname:'transparent'}}>
32             </div>
33             {isValid && colorname !== '' && (
34                 <p>
35                     Background color is set to: {colorname}
36                 </p>
37             )}
38             {!isValid && colorname !== '' &&(
39                 <p style={{color:'red'}}>
40                     Invalid color! Please select a valid color.
41                 </p>
42             )}
43         </div>
44     );
45 }
46
47
48 export default ColorChanger
```

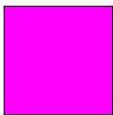
App.jsx:

```
App.jsx U x
Module 8 - Wk10 > Module 8_Week 10_Required Assignment_Patil > mytasks-app > src > App.jsx > ...
1  import { useState } from 'react'
2  import MessageButton from './Components/MessageButton'
3  import ColorChanger from './Components/ColorChanger'
4  import UserProfile from './Components/UserProfile'
5  import './App.css'
6
7  function App() {
8    const [count, setCount] = useState(0)
9
10   return (
11     <>
12       <h1>Task 1 - Handling Events in React</h1>
13       <div>
14         <MessageButton name="Alice" />
15       </div>
16       <h1>Task 2 - State Management using React Hooks</h1>
17       <div>
18         <ColorChanger/>
19       </div>
20       <h1>Task 3 - Asynchronous Data Fetching with AJAX</h1>
21       <div>
22         <UserProfile />
23       </div>
24     </>
25   )
26 }
27
28 export default App
29 |
```

The output: When color name is entered in the textbox, the background color for the below div is set automatically.

Task 2 - State Management using React Hooks

Enter the background color for the box:



Background color is set to: Magenta

Error Message:

Task 2 - State Management using React Hooks

Enter the background color for the box:

Invalid color! Please select a valid color.

Task 3. Asynchronous Data Fetching with AJAX

- a) Create a UserProfile component that fetches user data from an API (for example, <https://jsonplaceholder.typicode.com/users/1>) and displays the user's name, email, and address on the page.
- b) Display a loading message while data is being fetched.

Please find the screenshots for the UserProfile component.

- This component has a input box for the user to key in the User ID (type is set as number).
- When the user keys in valid User ID, the profile information for that user is fetched from 'https://jsonplaceholder.typicode.com/users' API and displayed in profile information section automatically.
- While the data is being fetched, the loading message is displayed in profile information section.
- If there is no User ID, entered in the input box, the profile information section displays, 'No data to display' message.
- When invalid User ID is entered, or any error occurs, the error message is displayed in the profile information section.
- This component is called from App.jsx file.

UserProfile.jsx

```
UserProfile.jsx U x
Module 8 - Wk10 > Module 8_Week 10_Required Assignment_Patil > mytasks-app > src > Components > UserProfile.jsx > UserProfile

1  import React, { useEffect, useState } from "react";
2
3  function UserProfile(){
4      const[userid, setUserId]=useState('')
5      const[showloading, setShowLoading]=useState(false)
6      const[showerror, setShowError]=useState(false)
7      const[userinfo, setUserInfo]=useState(null)
8
9      useEffect(() => {
10         /**** Check for valid User ID */
11         if (!userid) return;
12
13         /**** Set the loading message */
14         setShowLoading(true)
15         /**** Initialize error to false */
16         setShowError(false)
17         /**** Initialize userInfo to null */
18         setUserInfo(null)
19
20         fetch(`https://jsonplaceholder.typicode.com/users/${userid}`)
21         .then((response) => {
22             if (!response.ok) throw new Error('User not found');
23             return response.json();
24         })
25         .then((json) => setUserInfo(json))
26         .catch((err) => {
27             /**** Remove the loading message and show error message if any error */
28             setShowLoading(false)
29             setShowError(true)
30         })
31         .finally(() => {
32             /**** Remove the loading message */
33             setShowLoading(false)
34         })
35     }
36
37     },[userid]);
```

The code continues in the below screenshot...


```

38
39   return(
40     <div style={{padding:'20px'}}>
41       <h2>User Profile</h2>
42
43       <div>
44         <label title="Enter User ID" htmlFor="txtUserID"
45           style={{fontWeight:"bold"}}>Enter User ID: </label>
46         <input type="number" id="txtUserID" value={userid}
47           className="textbox" placeholder="Enter User ID to see the profile info"
48           onChange={(e) => setUserId(e.target.value)}></input>
49       </div>
50       <div style={{paddingTop:"20px"}}>
51         <hr/>
52         <h3>Profile Information:</h3>
53
54         {userinfo && (
55           <div>
56             <p><b>Name:</b> {userinfo.name}</p>
57             <p><b>User Name:</b> {userinfo.username}</p>
58             <p><b>Email:</b> {userinfo.email}</p>
59             <p><b>Address:</b> {userinfo.address.street}, {userinfo.address.suite}, {userinfo.address.city}, {userinfo.address.zipcode}</p>
60           </div>
61         )}
62         {userinfo === null && !showerror && !showloading && (
63           <div>
64             No data to display
65           </div>
66         )}
67         {showerror && (
68           <div>
69             <p style={{ color: 'red' }}>The error encountered. Please try again with a valid User ID.</p>
70           </div>
71         )}
72         {showloading && (
73           <div>
74             <p>Loading the user information! Please wait...</p>
75           </div>
76         )}
77       </div>
78     </div>
79   )
80
81
82
83
84   export default UserProfile

```

App.jsx:

```

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22          <UserProfile />
23        </div>
24      </>
25    )
26  }
27
28  export default App
29

```


Data fetched from API for valid User IDs:

Task 3 - Asynchronous Data Fetching with AJAX

User Profile

Enter User ID:

Profile Information:

Name: Leanne Graham

User Name: Bret

Email: Sincere@april.biz

Address: Kulas Light, Apt. 556, Gwenborough, 92998-3874

Task 3 - Asynchronous Data Fetching with AJAX

User Profile

Enter User ID:

Profile Information:

Name: Clementina DuBuque

User Name: Moriah.Stanton

Email: Rey.Padberg@karina.biz

Address: Kattie Turnpike, Suite 198, Lebsackbury, 31428-2261

Error:

Task 3 - Asynchronous Data Fetching with AJAX

User Profile

Enter User ID:

Profile Information:

The error encountered. Please try again with a valid User ID.